Akoizo Original Practice - Form 2

Division B | Botany | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Which tissue primarily carries water and minerals?

Unit: Plant cells, tissues, and vascular organization.
Worked-case data: A mature leaf exports sugar to a growing root. Identify source, sink, and transport tissue.

A. Epidermal hairs only

B. Pollen

C. Xylem

D. Phloem only

Answer: _____

2. (2 points) A plant spore is produced by…

Unit: Life cycles, flowers, seeds, and biotechnology.
Worked-case data: Track ploidy in the standard angiosperm pathway: diploid sporophyte, spore, gametophyte, gamete, zygote.

A. Mitosis of a diploid egg only

B. Pollination alone

C. Meiosis in the sporophyte

D. Fertilization of two seeds

Answer: _____

3. (2 points) Rotating branches at a node changes relationships?

Unit: Plant evolution, fossils, and diversity.
Worked-case data: Two modern lineages share a vascular-tissue ancestor. Does one living lineage have to be the ancestor of the other?

A. Only for plants

B. Only for fossils

C. No

D. Yes

Answer: _____

4. (2 points) Yellow leaves uniquely prove nitrogen deficiency?

Unit: Botany, evidence, and ecosystem importance.
Worked-case data: A plant community fixes 100 carbon units and its producers respire 40 units over the same interval. Calculate NPP.

A. Yes

B. Only in all monocots

C. Only at noon

D. No

Answer: _____

5. (2 points) Tall-tree xylem flow is generally explained by…

Unit: Water transport, storage, hormones, and growth.
Worked-case data: Soil water potential is -0.2 MPa, root -0.5 MPa, leaf -1.0 MPa. Predict the idealized direction.

A. Each vessel pumping with ATP

B. Phloem always pushing upward

C. Gravity pulling water upward

D. Cohesion-tension linked to transpiration

Answer: _____

6. (4 points) Explain the evidence supporting your answer, including the calculation or comparison.

Unit: Photosynthesis and plant energy.
Worked-case data: Low light: 5 bubbles per minute. Medium light: 10 bubbles per minute. High light: 10 bubbles per minute at the same temperature and carbon-dioxide supply.
Separate investigation observations:
Low light: 5 bubbles per minute.
Medium light: 10 bubbles per minute.
High light: 10 bubbles per minute at the same temperature and carbon-dioxide supply.

Answer: _____
_____
_____

7. (2 points) Natural origin guarantees medicinal safety?

Unit: Horticulture, aquatic systems, medicine, and materials.
Worked-case data: Two irrigation treatments produce the same growth index, but one uses twice the water. What else is needed to choose a management approach?

A. Yes

B. Only for leaves

C. Only for roots

D. No

Answer: _____

8. (2 points) A thicker cuticle commonly helps reduce…

Unit: Roots, stems, and leaves.
Worked-case data: A leaf closes its stomata during dry conditions. Predict one benefit and one cost.

A. Uncontrolled water loss

B. All respiration

C. Every nutrient deficiency

D. All need for roots

Answer: _____

9. (2 points) “Cryptogams” is best treated as…

Unit: Major plant groups and comparative identification.
Worked-case data: A specimen has parallel leaf venation, floral parts in threes, and scattered stem bundles. Identify the supported broad group.

A. A historical descriptive grouping

B. One exclusive modern clade

C. A synonym for all angiosperms

D. A seed tissue

Answer: _____

10. (2 points) Changing water and nutrients together reveals the unique cause?

Unit: Plant ecology, nutrient deficiencies, and disease.
Worked-case data: Older leaves yellow first, but soil is dry and no tissue analysis exists. Is nutrient deficiency established?

A. No

B. Yes

C. Only if leaves become greener

D. Only at high light

Answer: _____

11. (4 points) Distinguish wall and membrane.

Unit: Plant cells, tissues, and vascular organization.
Worked-case data: A mature leaf exports sugar to a growing root. Identify source, sink, and transport tissue.

Answer: _____
_____
_____

12. (4 points) Track n and 2n in the example.

Answer: _____
_____
_____

13. (4 points) Why is a tree not a progress ladder?

Unit: Plant evolution, fossils, and diversity.
Worked-case data: Two modern lineages share a vascular-tissue ancestor. Does one living lineage have to be the ancestor of the other?

Answer: _____
_____
_____

14. (4 points) Why do plants need respiration?

Unit: Botany, evidence, and ecosystem importance.
Worked-case data: A plant community fixes 100 carbon units and its producers respire 40 units over the same interval. Calculate NPP.

Answer: _____
_____
_____

15. (4 points) Why distinguish phloem and xylem mechanisms?

Unit: Water transport, storage, hormones, and growth.
Worked-case data: Soil water potential is -0.2 MPa, root -0.5 MPa, leaf -1.0 MPa. Predict the idealized direction.

Answer: _____
_____
_____

16. (4 points) Name a terrestrial tradeoff.

Unit: Plant evolution, fossils, and diversity.
Worked-case data: Two modern lineages share a vascular-tissue ancestor. Does one living lineage have to be the ancestor of the other?

Answer: _____
_____
_____

17. (4 points) Name two plant ecosystem functions.

Unit: Botany, evidence, and ecosystem importance.
Worked-case data: A plant community fixes 100 carbon units and its producers respire 40 units over the same interval. Calculate NPP.

Answer: _____
_____
_____

18. (4 points) What does a directional-light control test?

Unit: Water transport, storage, hormones, and growth.
Worked-case data: Soil water potential is -0.2 MPa, root -0.5 MPa, leaf -1.0 MPa. Predict the idealized direction.

Answer: _____
_____
_____